

Serie 18

CHARACTERISTIC POLYNOMIALS, EIGENVECTORS, EIGENVALUES

1. Consider the matrix $A = \begin{pmatrix} 3 & 0 & -2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$ over $\mathbb{R}$.
 - (a) Determine the characteristic polynomial of A .
 - (b) Determine the eigenvalues of A .
 - (c) The *geometric multiplicity* of an eigenvector is the dimension of its eigenspace. The *arithmetic multiplicity* of an eigenvector is the multiplicity of this eigenvector as a zero of the characteristic polynomial. Determine the arithmetic and geometric multiplicity of all eigenvalues.
2. Compute the characteristic polynomial, the eigenvalues and eigenvectors of the following matrices over $\mathbb{Q}$ and check if they are diagonalizable.
 - (a) $A := \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix}$
 - (b) $B := \begin{pmatrix} 2 & 2 & 3 \\ 1 & 2 & 1 \\ 2 & -2 & 1 \end{pmatrix}$
 - (c) $C := \begin{pmatrix} -4 & -3 & -1 & -7 \\ -3 & -1 & -1 & -4 \\ 6 & 4 & 3 & 8 \\ 3 & 3 & 1 & 6 \end{pmatrix}$
3. For an arbitrary invertible $n \times n$ -matrix A , write the characteristic polynomial of A^{-1} in terms of the characteristic polynomial of A .
4. Let K^∞ be the vectorspace of all infinite sequences in K , and let K_0^∞ be the subspace of all sequences where all but finitely many elements are 0.
 - (a) Determine all eigenvalues and eigenvectors of the endomorphism
$$T : K^\infty \rightarrow K^\infty, (x_0, x_1, x_2, \dots) \mapsto (x_1, x_2, x_3, \dots).$$
 - (b) Do the same for the induced endomorphism $K_0^\infty \rightarrow K_0^\infty$.
 - (c) Construct an endomorphism of K_0^∞ with eigenvalues $0, 1, 2, 3, \dots$

- (d) Construct an endomorphism of K_0^∞ which has no Eigenvalues.
5. Let A be a nilpotent $n \times n$ real matrix. This means that there exists $m \geq 1$ with $A^m = O$. Show that the only possible eigenvalue of A is 0. When exactly is 0 an eigenvalue of A ?
6. Let V be a K -vectorspace and let $F, G \in \text{End}(V)$. Show:
- (a) If $v \in V$ is an eigenvector of $F \circ G$ with eigenvalue λ and $G(v) \neq 0$, then $G(v)$ is an eigenvector of $G \circ F$ with eigenvalue λ .
 - (b) If V is finite-dimensional, the endomorphisms $F \circ G$ and $G \circ F$ have the same eigenvalues.
 - (c) Find a counterexample to (b) if V is not finite-dimensional.
7. Let **Fib** be the set of all sequences $(a_0, a_1, a_2, \dots, a_n, \dots) = \{a_i\}_{i=0}^\infty$ in $\mathbb{R}^\infty$ that satisfy $a_n = a_{n-1} + a_{n-2}$ for all $n \geq 2$. Show:
- (a) The map $F : \mathbf{Fib} \rightarrow \mathbb{R}^2, F(\{a_i\}_{i=0}^\infty) := (a_0, a_1)$ is an isomorphism.
 - (b) Consider the map shifting the sequence $T : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty, (a_0, a_1, a_2, \dots) \mapsto (a_1, a_2, a_3, \dots)$. Find all eigenvectors and eigenvalues of this map. Show that T sends **Fib** to **Fib**, and find all eigenvalues and eigenvectors of $T|_{\mathbf{Fib}} : \mathbf{Fib} \rightarrow \mathbf{Fib}$.
 - (c) Show that the eigenvectors of $T|_{\mathbf{Fib}}$ from part (7b) form a basis for **Fib**.